

Detecting and Understanding Electricity Price Anomalies in Queensland's NEM Market

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IFN695 Assessment 3

Executive Summary

Electricity price anomalies within Queensland's National Electricity Market (NEM) are infrequent but carry significant operational and financial consequences. These anomalies are typically triggered by abrupt changes in demand, generation shortfalls, market forecast errors, or extreme weather events. Traditional systems, which often rely on pre-defined thresholds or labeled datasets, struggle to detect such anomalies in real time—especially in the dynamic and high-frequency environment of electricity markets.

To address these challenges, this project explores a dual-model, unsupervised learning framework using **Isolation Forest** and **LSTM Autoencoder** to detect electricity price anomalies without the need for ground truth labels. Historical data was collected from two authoritative sources: the **Australian Energy Market Operator (AEMO)**, which provided market pricing and operational metrics, and the **Bureau of Meteorology (BoM)**, which supplied environmental observations such as temperature, solar radiation, and rainfall. These datasets were integrated and filtered to focus exclusively on the Queensland region from January 2022 to December 2024, resulting in approximately 500,000 records at a 5-minute granularity.

Key features such as lagged RRP values, percentage price change, demand error (difference between forecast and actual demand), and time-based attributes were engineered to capture temporal and contextual factors influencing price volatility. These features were normalized and used as input for both models.

The **Isolation Forest** algorithm was employed to identify feature-space outliers, while the **LSTM Autoencoder** focused on capturing temporal deviations in electricity price sequences. Anomalies detected by both models were considered high-confidence events. The results showed that although the two models captured different types of irregularities, their combined use improved anomaly detection reliability.

Several meaningful patterns were identified. Anomalies often aligned with high ambient temperatures, sharp deviations in demand forecasts, and occurred disproportionately during weekday mornings and weekend evenings—periods typically associated with grid stress. These insights not only validate the feature choices but also reinforce the operational value of unsupervised anomaly detection in electricity markets.

In summary, this project delivers a scalable, interpretable, and label-free framework for detecting electricity price anomalies. It demonstrates the potential for integration into real-time market surveillance systems and lays the foundation for future enhancements such as anomaly classification and dashboard-based visualization tools.

Introduction

Volatility in electricity prices poses a significant operational and financial challenge across global energy markets, with Australia's National Electricity Market (NEM) being no exception. In Queensland, rapid and unforeseen price spikes—often driven by demand surges, forecast inaccuracies, supply disruptions, or extreme weather events—can disrupt market stability and lead to substantial economic consequences for energy providers, regulators, and consumers. The ability to detect these anomalies early is essential for informed decision-making, risk management, and maintaining the reliability of energy supply.

However, traditional anomaly detection methods in electricity markets are often reliant on static thresholds or supervised learning models that require labeled historical anomalies. This limits their flexibility, especially in real-time applications where patterns are non-linear, contextual, and frequently evolving. To address this limitation, the current project employs an unsupervised machine learning framework that eliminates the dependency on predefined labels and adapts to dynamic market behaviour.

The central goal of this study is to develop a robust, data-driven system capable of detecting anomalous electricity price behaviour using a combination of market and environmental data from 2022 to 2024. The system integrates historical dispatch and trading data from the Australian Energy Market Operator (AEMO) with meteorological observations from the Bureau of Meteorology (BoM). Through preprocessing, feature engineering, and model training, the study leverages both spatial and temporal patterns in the data to flag anomalies that would otherwise go unnoticed.

Two core models are applied: **Isolation Forest**, which identifies outliers based on feature rarity in multi-dimensional space, and **LSTM Autoencoder**, which reconstructs normal time-series sequences and flags deviations based on reconstruction error. Anomalies confirmed by both models are treated as high-confidence events, increasing the credibility of detection.

This report outlines the background research motivating the approach, details the methodology and preprocessing pipeline, and presents key findings supported by visual analysis. It concludes with a reflection on limitations and potential future directions to enhance the system's performance and operational relevance.

Literature Review

Electricity price anomaly detection has become increasingly important, especially in deregulated and volatile markets such as Australia's National Electricity Market (NEM). Traditional detection methods primarily rely on rule-based systems or supervised learning models that require labeled anomaly data, which are often limited or unavailable. To address these limitations, recent studies have turned

toward unsupervised and hybrid machine learning approaches for more scalable and adaptable anomaly detection [1], [2].

Mohammadi et al. [1] present a comprehensive review of machine learning applications in electricity markets, emphasizing the shift towards data-driven methods for tasks such as price forecasting, bidding strategies, and anomaly detection. Their findings highlight the growing adoption of unsupervised models due to the scarcity of labelled training data in real-world scenarios.

Environmental factors also significantly influence electricity price behaviour. Sgarlato and Ziel [7] demonstrate that incorporating weather forecasts—particularly temperature, solar radiation, and rainfall—can enhance electricity price modelling, even beyond the day-ahead market. Their work supports the inclusion of exogenous weather variables from the Bureau of Meteorology in anomaly detection systems.

Focusing on time-series anomaly detection, LSTM Autoencoders have emerged as effective tools for learning complex temporal patterns in electricity usage. Ghanim and Awad [3] proposed a reinforcement learning and time series forest-based framework for unsupervised anomaly detection in electricity consumption, demonstrating the adaptability of hybrid techniques. Additionally, LSTM Autoencoder-based models have been shown to capture reconstruction errors in time-series data, even under noise or periodicity, as illustrated by studies such as Shrestha et al. [4], and Homayouni et al. [2].

Isolation Forest is another well-established anomaly detection technique that has been successfully applied in multivariate electricity datasets. Introduced by Liu et al. [6], the algorithm isolates rare instances through recursive partitioning, offering a computationally efficient and scalable method. It has since been adopted in smart grid contexts and energy monitoring systems due to its simplicity and effectiveness.

Together, these studies validate the methodological choices in this project—combining the temporal detection strength of LSTM Autoencoders with the spatial outlier detection capabilities of Isolation Forest, enhanced by weather features. This integrated approach extends current literature by applying both models to high-frequency electricity pricing and meteorological data from Queensland, with the aim of uncovering patterns and high-confidence anomalies linked to operational and climatic factors.

Methodology

The core aim of this project was to detect and analyze electricity price anomalies in Queensland's National Electricity Market (NEM) using unsupervised machine learning techniques. To systematically achieve this, a five-stage methodology was developed comprising: (1) data collection, (2) data preprocessing, (3) feature engineering, (4) anomaly detection using machine learning models, and (5) model evaluation using visual analytics.

Data Collection

The dataset was compiled by integrating high-frequency market and weather datasets from two primary sources: the Australian Energy Market Operator (AEMO) and the Bureau of Meteorology (BoM). Three datasets from AEMO were used:

- **DISPATCHPRICE:** Offered 5-minute interval data capturing the Regional Reference Price (RRP), which is the central variable in this study. It also included ancillary service prices like RAISE6SECRRP and LOWER5MINRRP, although only RRP was retained for modeling.
- **DISPATCHREGIONSUM:** Contained operational metrics critical for understanding supply-demand dynamics, such as TOTALDEMAND, DEMANDFORECAST, AVAILABLEGENERATION, and DISPATCHABLEGENERATION.
- **TRADINGPRICE:** Provided a 30-minute aggregated RRP value known as TRADING_RRP, used to study short-term price smoothing and stability.

From BoM, hourly environmental observations were acquired for Queensland. These included:

- **TEMPERATURE**
- **RAINFALL**
- **SOLAR_RADIATION**

All datasets were merged using the SETTLEMENTDATE field as the common time key, after appropriate frequency alignment and regional filtering (QLD). The resulting merged dataset comprised approximately 500,000 rows of data at 5-minute granularity over a multi-year period.

Data Preprocessing

Before training the models, extensive preprocessing steps were undertaken to ensure data quality and consistency:

- **Missing Values & Placeholder Handling:** Certain values, such as -999.99 in RRP or environmental variables, were identified as placeholders and replaced with NaN. A two-pass filling method—forward fill followed by backward fill—was applied to interpolate these gaps while preserving temporal integrity.
- **Datetime Conversion:** The SETTLEMENTDATE column was transformed into a Python datetime object. This allowed for easier time-based feature creation and alignment across datasets.
- **Scaling:** All numerical features were normalized to a [0, 1] range using MinMaxScaler. This step was essential for neural network stability, especially for the LSTM Autoencoder, as differing feature scales can skew training.

Feature Engineering

To enable both temporal and contextual anomaly detection, a set of derived features were constructed from the raw data:

Feature Name	Description
lagged_RRP	Previous 5-minute RRP value (t-1), used to assess price momentum.
price_change_pct	Percentage change in RRP over the last interval to capture volatility.
demand_error	Difference between TOTALDEMAND and DEMANDFORECAST to capture forecast inaccuracy.
hour_of_day	Integer representing the hour (0–23), capturing intraday patterns.
day_of_week	Integer (0–6) encoding day of the week, useful for detecting weekly cycles.

These features were selected based on literature insights and domain relevance for capturing market behavior and volatility.

Anomaly Detection Models

Two unsupervised models were employed to detect anomalies in different ways—Isolation Forest (IF) for spatial outliers and LSTM Autoencoder for temporal deviations.

Isolation Forest

- **Objective:** Identify feature-space anomalies where records are isolated from typical data clusters.
- **Mechanism:** Constructs random binary trees and isolates data points that require fewer splits to separate.
- **Hyperparameters:**
 - contamination = 0.02: Indicates 2% of the data is assumed to be anomalous.
 - n_estimators = 100: Number of trees in the forest.
- **Output:** A binary anomaly score (1 = anomaly, 0 = normal), saved as a new column for evaluation.

LSTM Autoencoder

- **Objective:** Detect sequences of electricity prices and contextual data that deviate from learned patterns.
- **Architecture:**
 - Input: 96 time steps (equivalent to 8 hours).
 - Layers: LSTM Encoder → Dense Bottleneck → LSTM Decoder.
 - Output: Reconstructed sequence of the same dimension as the input.

- **Training:** Trained to minimize Mean Squared Error between input and reconstruction.
- **Anomaly Scoring:** Instances with reconstruction error above the 95th percentile were labeled as anomalies.

Evaluation and Labelling Strategy

Since no labelled anomalies exist in the dataset, conventional supervised metrics (accuracy, precision) were not applicable. The evaluation was based on the following strategies:

1. Dual-Model Agreement

To reduce false positives, high-confidence anomalies were defined as data points concurrently flagged by both Isolation Forest and LSTM Autoencoder. A new boolean column, `both_agree`, was added to highlight such overlapping detections.

2. Visual Analysis

To interpret and validate anomaly patterns, multiple visual techniques were employed:

- **Venn Diagram:** Illustrated the volume of anomalies detected by each model and their intersection.
- **Time Series Overlay:** Plotted RRP with anomaly flags from both models to show when and how spikes aligned with model predictions.
- **Heatmap (Day vs Hour):** Revealed temporal hotspots for anomalies—useful for identifying operational cycles.
- **Scatter Plot (RRP vs. Temperature):** Explored the relationship between climatic extremes and pricing anomalies.
- **Bar Chart (Top 10 Errors):** Showcased the most severe anomaly scores from the LSTM Autoencoder.
- **Radar/Deviation Chart:** Compared median feature values between anomalous and normal instances to explain driving factors.

Results and Findings

This section presents the key outputs from both models used—Isolation Forest and LSTM Autoencoder—along with quantitative results and accompanying visualizations to highlight temporal patterns, model complementarity, and anomaly characteristics.

1. Anomaly Counts

The LSTM Autoencoder flagged approximately **5%** of the dataset (9,289 records) as anomalous using a reconstruction error threshold at the 95th percentile. Meanwhile, the Isolation Forest model identified about **2%** of the data (2,780 records) as anomalies using a contamination rate of 0.02. A

total of **1,563 data points** (~0.7%) were detected by both models, which were classified as **high-confidence anomalies**.

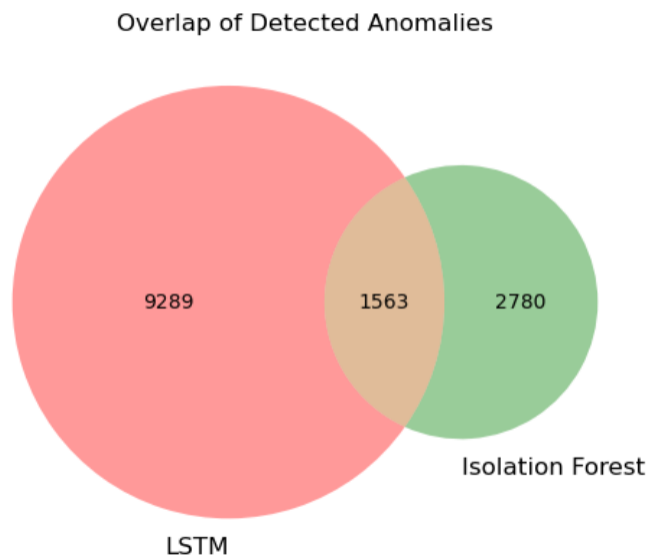


Figure 1: Venn Diagram Showing Overlap Between Anomalies Detected by LSTM and Isolation Forest

Figure 1 illustrates the overlap between the LSTM and Isolation Forest models. While each model detects its own set of anomalies, only a small portion (~0.7%) of anomalies are jointly identified. This limited intersection reflects the **complementary strengths** of both models—Isolation Forest focuses on detecting feature-based outliers (e.g., demand spikes), while LSTM Autoencoder captures **temporal anomalies**, such as unusual sequential price behavior. The combination of both models enhances overall detection confidence and coverage.

2. Temporal Patterns

The LSTM Autoencoder revealed clear temporal clustering in anomaly occurrences. Anomalies were most frequently observed during **weekday mornings** and **weekend evenings**—time periods that align with routine **peak demand** and transitional phases in grid load balancing. These patterns suggest a strong link between electricity price anomalies and operational cycles within the National Electricity Market (NEM).

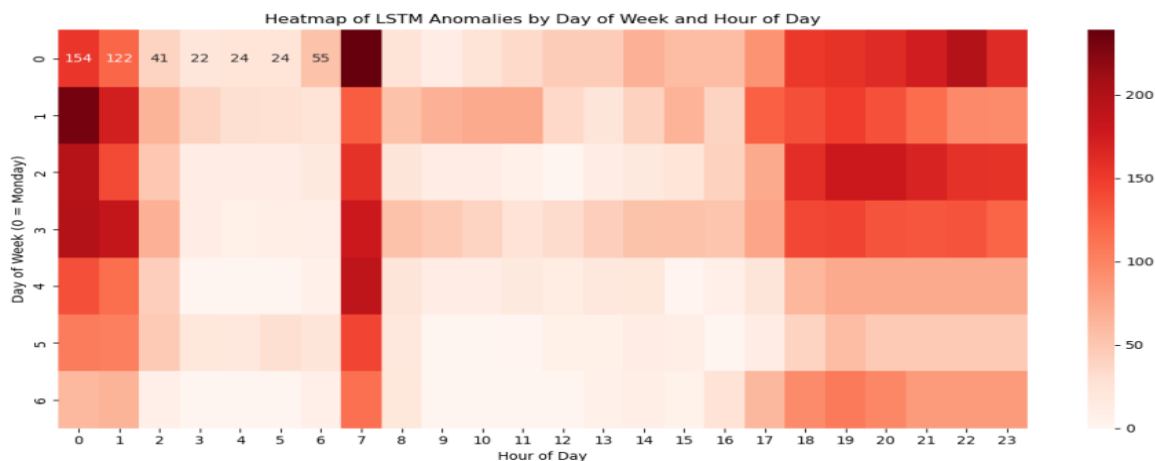


Figure 2: Heatmap of LSTM Anomalies by Hour of Day and Day of Week

Figure 2 highlights how LSTM-detected anomalies are concentrated during specific time windows—most notably on **weekday mornings** and **late weekends**. These timeframes are known for **high demand volatility**, often associated with changes in consumer usage patterns and generator dispatch schedules. The heatmap confirms that electricity price spikes tend to follow **structural stress points** in the grid, validating the importance of time-based features like `hour_of_day` and `day_of_week` in anomaly detection models.

3. Anomaly Severity

The top 10 most severe anomalies were identified using reconstruction error from the LSTM Autoencoder. These instances included both **unrealistic placeholder values** (e.g., -999.99) and legitimate **market spikes**, with Regional Reference Prices reaching extreme values such as **\$15,100/MWh**. These cases were considered the most critical anomalies from both a technical and economic standpoint.

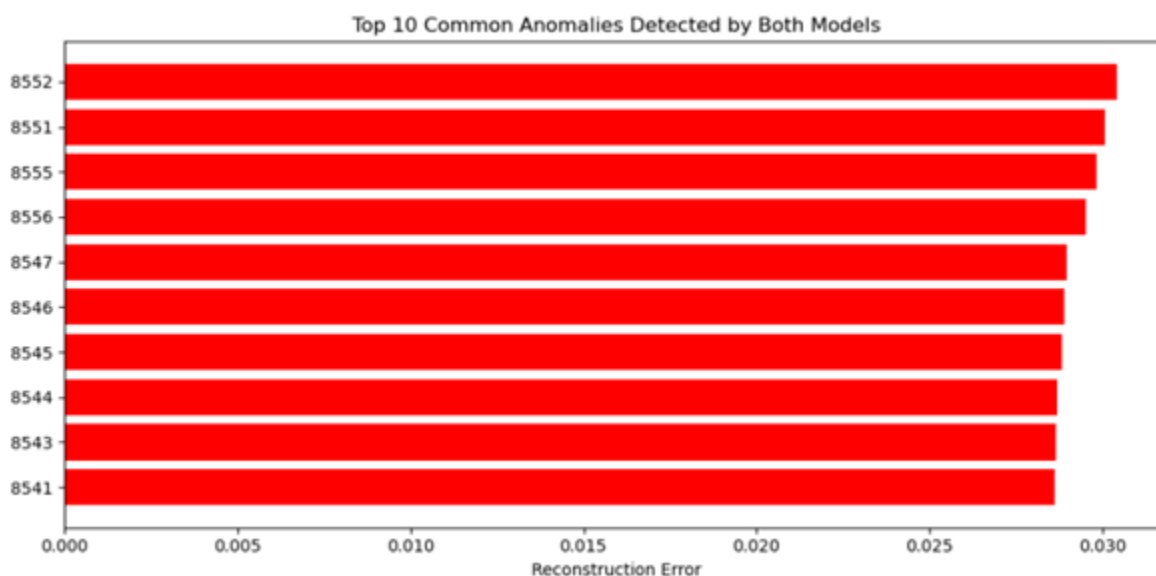


Figure 3: Top 10 Anomalies by Reconstruction Error

Figure 3 displays the top 10 high-confidence anomalies, sorted by reconstruction error. A closer look at these points revealed that several contained **invalid placeholders** or **price spikes well above the regulatory cap**. These anomalies not only indicate **data quality issues** but also represent real market disturbances that can impact grid planning and financial forecasting. This emphasizes the dual need for **automated anomaly detection** and **manual validation processes** in real-world electricity markets.

4. Environmental Influence

A strong relationship was observed between **temperature** and anomalous pricing events. The scatter plot comparing **RRP** with **ambient temperature** showed that many of the anomalies—especially those with extreme price spikes—occurred at **elevated temperature levels**, often exceeding **34°C**. This correlation suggests that **weather stress** plays a significant role in triggering price volatility within the grid.

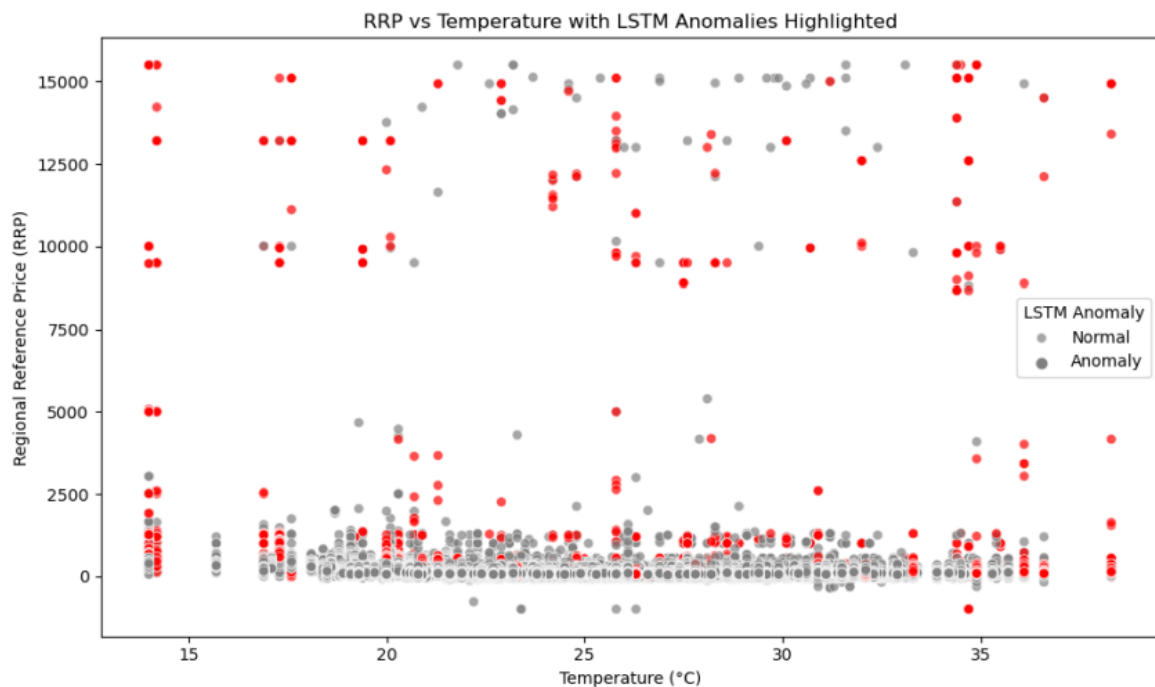


Figure 4: Scatter Plot of RRP vs. Temperature with LSTM Anomalies Highlighted

Figure 4 clearly illustrates that **anomalies predominantly cluster at higher temperature ranges**. This supports the hypothesis that extreme heat, often linked with increased electricity demand and supply strain, contributes significantly to electricity price anomalies. By highlighting LSTM-detected anomalies in red, the plot demonstrates the **effectiveness of using weather data**, particularly temperature, in anomaly detection models.

5. Feature Contribution

To interpret the underlying factors driving anomalies, a **median deviation analysis** was performed between anomalous and normal instances. Features such as **TOTALDEMAND**, **demand_error**, and **DISPATCHABLEGENERATION** showed the highest deviation values, suggesting they contributed most to the model's anomaly detection. In contrast, environmental features like **TEMPERATURE**,

SOLAR_RADIATION, and **RAINFALL** showed relatively minor deviations, indicating a more indirect influence.

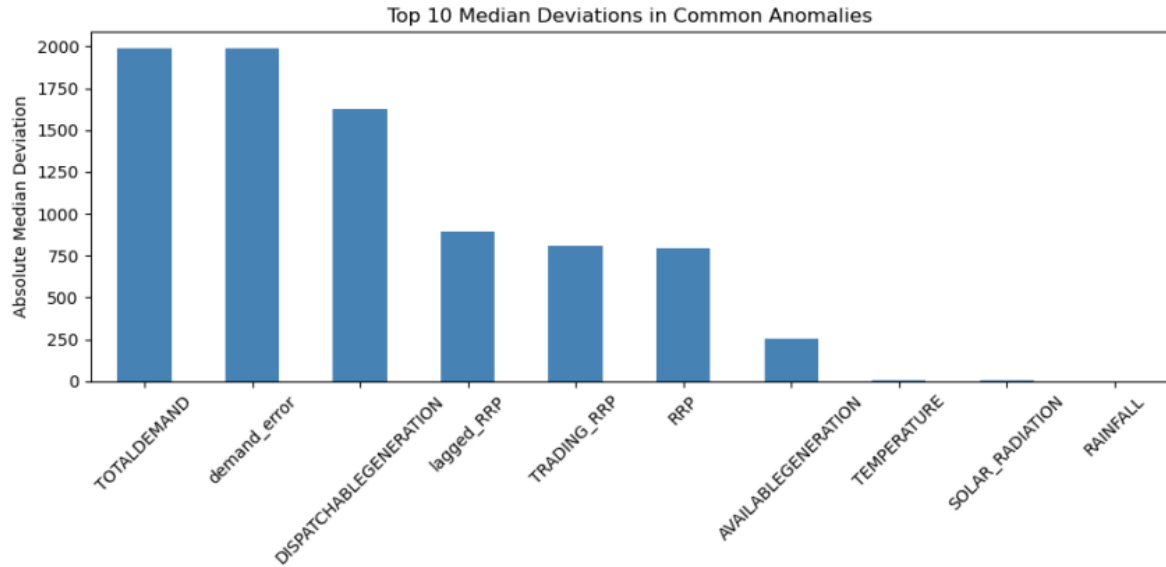


Figure 5: Top 10 Median Deviations in Common Anomalies

Figure 5 reveals that **demand-related variables** were the most influential in separating anomalous events from normal behavior. Notably, **TOTALDEMAND** and **demand_error** had the largest deviations, confirming that mismatches between expected and actual demand are key drivers of electricity price anomalies. **DISPATCHABLEGENERATION** also played a strong role, indicating that supply-side limitations significantly affect market volatility. These findings validate the inclusion of these features in the model and offer interpretable evidence of their importance.

6. Combined Visual Timeline

To analyze anomaly unfolding over time, a **time-series overlay** of the Regional Reference Price (RRP) was created, highlighting points where the **LSTM Autoencoder**, **Isolation Forest**, or **both models** flagged anomalies. This visual context shows how anomalies align with extreme price peaks and whether they were isolated or captured by both methods simultaneously.

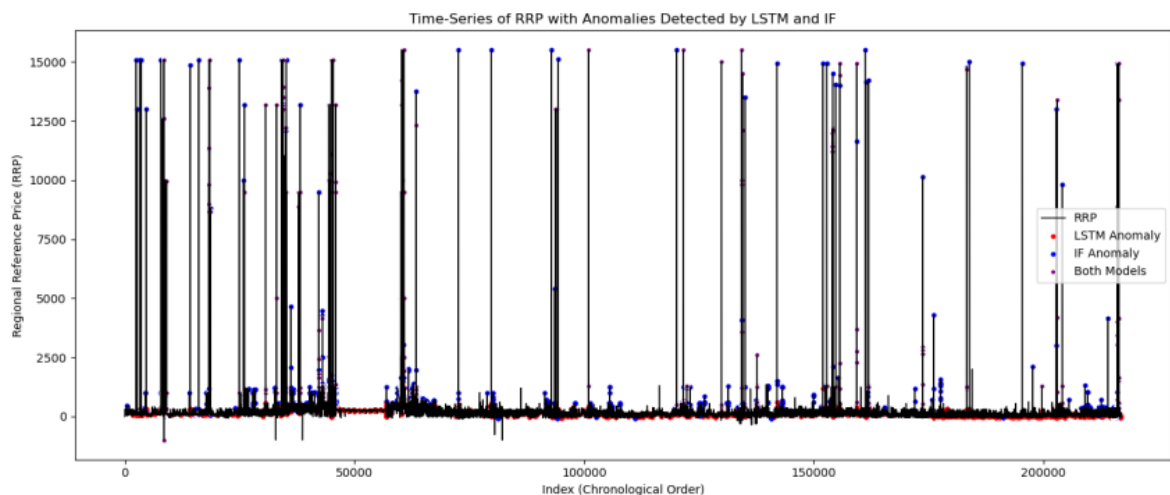


Figure 6: Time-Series of RRP with Anomalies Detected by LSTM and IF

Figure 6 offers a comprehensive view of model behavior across the timeline. It confirms that **LSTM-detected anomalies (red)** are distributed more broadly across fluctuations, while **IF anomalies (blue)** are more sparsely concentrated around sharper spikes. **Purple markers**, representing agreement between both models, appear during the most extreme price disruptions. This demonstrates how a **dual-model strategy provides layered detection**—capturing both subtle and pronounced deviations—and reinforces the importance of using both temporal and spatial perspectives in anomaly analysis.

Significance and Limitations

Significance

This project establishes a robust, flexible, and scalable framework for detecting electricity price anomalies in Queensland's National Electricity Market (NEM) using unsupervised learning methods. A notable strength of this approach lies in its independence from labeled training data, which is typically scarce or unavailable in electricity market operations. By leveraging algorithms such as Isolation Forest and LSTM Autoencoder, the model effectively identifies irregularities in both feature space and temporal sequences, thus adapting well to the volatile and rapidly changing nature of electricity pricing.

A major accomplishment of this work is its successful implementation of label-free anomaly detection. The Isolation Forest model isolates rare instances based on decision path lengths, while the LSTM Autoencoder learns to reconstruct normal sequences and flags deviations with high reconstruction error. This enables detection of unexpected pricing behaviors without relying on prior annotations or historical anomaly logs, which makes the solution particularly relevant for real-time grid monitoring where unknown or emerging patterns may occur.

Furthermore, the integration of two complementary models significantly enhances the reliability of anomaly detection. The dual-model validation strategy—where anomalies flagged by both models are considered high-confidence—minimizes the risk of false alarms and increases the interpretability of results. These overlapping detections often coincided with well-known stress events such as extreme demand spikes or weather-induced volatility, suggesting practical applicability in operational environments.

The methodology also demonstrated strong operational relevance. Anomalies detected through this framework were not just theoretical artifacts but corresponded with real-world conditions, such as peak load periods or extreme temperatures. This alignment supports potential use cases in energy trading, risk forecasting, or grid intervention planning. Moreover, the methodology's modular structure ensures scalability beyond the current case study. The same architecture can be applied to other NEM regions like New South Wales or Victoria, or adapted for new market scenarios involving battery storage, demand response programs, or renewable integration.

Limitations

While the project offers a promising and practical solution for electricity price anomaly detection, several limitations were identified. A key challenge was the prevalence of false positives, particularly from the Isolation Forest model. As an unsupervised method, it occasionally flagged points that may be statistically anomalous but lack real-world significance, highlighting the need for contextual or rule-based post-processing in operational settings. Without such domain-based filtering, there is a risk of overwhelming analysts with non-actionable detections.

Another limitation stems from the complexity of the LSTM Autoencoder. While it provided valuable insights into temporal anomalies, the model required substantial computational resources and fine-tuning. Sequence length, batch size, number of layers, and threshold sensitivity all had to be calibrated carefully to achieve meaningful results. This added overhead can pose constraints in real-time deployment or when working with larger datasets unless architectural optimizations are applied.

The project was also constrained by the lack of ground truth labels. In the absence of predefined anomaly annotations, it was not possible to use conventional evaluation metrics such as precision, recall, or F1-score. As a result, model performance had to be evaluated through indirect methods, including visual analysis, anomaly count comparisons, and cross-model agreement. While this approach is reasonable for exploratory studies, it limits statistical rigor and makes benchmarking against other models more difficult.

Lastly, the scope of features used in the model was restricted by data availability. Key operational indicators—such as generator outage logs, real-time congestion metrics, or intervention notices—were not accessible, and thus could not be incorporated. Their inclusion could have improved model interpretability and helped explain certain anomalies that appeared ambiguous with the current feature set. Without these operational insights, some detections remained partially unexplained, reducing the explanatory power and utility of the system for grid operators.

In summary, while the framework developed in this project demonstrates strong adaptability and analytical value, future work should aim to address these limitations through enhanced feature integration, improved model efficiency, and the establishment of evaluation benchmarks using labeled datasets or expert-verified events.

Future Work

Building upon the results of this project, several areas for future enhancement and expansion have been identified to improve the effectiveness, scalability, and operational relevance of the anomaly detection framework.

Firstly, the current methodology can be extended to other regions within the National Electricity Market (NEM), such as New South Wales and Victoria. This would allow for cross-regional benchmarking of pricing anomalies and help assess the generalizability and adaptability of the dual-model detection strategy under varying market and climatic conditions.

Secondly, the inclusion of additional operational datasets is recommended to enhance the contextual interpretation of anomalies. Data such as generator outage reports, real-time congestion metrics, and market intervention notices could help distinguish between anomalies caused by

internal market behavior and those triggered by external operational events. This may lead to improved anomaly classification and a reduction in false positive rates.

Furthermore, the implementation of a real-time monitoring pipeline should be considered. By leveraging AEMO's live data APIs, the detection models could be adapted to provide continuous anomaly monitoring, enabling system operators and market analysts to act promptly on emerging volatility. To support real-time deployment, optimization of the LSTM Autoencoder is necessary—potentially through the use of more efficient architectures such as Gated Recurrent Units (GRUs) or Transformer-based models, and by applying model compression techniques like pruning or quantization.

Additionally, a post-detection classification stage could be introduced to categorize identified anomalies by their underlying drivers, such as demand surges, weather extremes, or technical constraints. This would enhance the interpretability of detected events and facilitate better decision-making for risk mitigation.

Finally, the development of an interactive visualization dashboard is proposed. Tools like Plotly Dash or Streamlit could be employed to create a platform for analysts to explore anomalies alongside key contextual data in real time. Such a system would provide operational value by making anomaly detection results more accessible and actionable for stakeholders.

These future enhancements aim to make the framework more robust, interpretable, and operationally deployable across the energy sector.

Reflection

This project provided a comprehensive learning experience in applying unsupervised machine learning to real-world time-series data, particularly in the energy domain. Working with electricity market data from AEMO and weather data from the Bureau of Meteorology offered a practical understanding of the complexities involved in integrating, aligning, and analyzing multi-source datasets.

A key takeaway was the implementation of a dual-model anomaly detection strategy using Isolation Forest and LSTM Autoencoder. Isolation Forest was relatively straightforward, offering quick insights into outlier detection in the feature space. In contrast, the LSTM Autoencoder demanded deeper engagement—requiring careful attention to sequence structuring, input scaling, and threshold tuning. This contrast between model simplicity and depth helped me appreciate the trade-offs between speed and temporal sensitivity.

The project also highlighted the importance of visualization not just as a presentation tool, but as a core analytical method. Tools like time-series overlays, heatmaps, and deviation plots played a vital role in interpreting model behaviour and understanding when and why anomalies occurred. These visuals added believability to the results and guided deeper exploration of underlying patterns.

Another valuable insight came from comparing model outputs. While overlapping anomalies provided high-confidence signals, the differences between LSTM and IF flagged the importance of

context—temporal versus static detection. This raised thoughtful questions about how best to interpret anomalies and whether further classification is needed in future work.

Overall, this project enhanced my understanding of unsupervised learning, anomaly detection, and time-series modeling. It also strengthened my ability to work across disciplines, from energy systems to machine learning, and to present technical findings in an interpretable, impactful way.

Conclusion

This project developed and validated an unsupervised, dual-model framework for detecting electricity price anomalies in Queensland's National Electricity Market (NEM). By integrating high-resolution market data from AEMO with weather observations from the Bureau of Meteorology, the approach successfully captured both operational and environmental contributors to price volatility. The use of Isolation Forest for spatial anomaly detection and LSTM Autoencoder for temporal sequence deviation provided a comprehensive view of abnormal price behaviour, with overlapping detections offering high-confidence signals.

Key insights from the analysis included the temporal clustering of anomalies during grid stress periods, such as weekday mornings and weekend evenings, and a strong association between high temperatures and extreme price spikes. These findings validate the methodological choices and highlight the potential of unsupervised learning for real-time market surveillance, especially in data-constrained environments where labelled anomalies are unavailable.

While the system demonstrated strong adaptability and interpretability, the study also acknowledged limitations such as the lack of ground truth labels, potential false positives, and the absence of some operational features like generator outage reports. Addressing these limitations through future integration of additional data sources, real-time model optimization, and post-detection classification techniques could further enhance the system's accuracy and relevance.

Ultimately, this project contributes a scalable and domain-aware approach to anomaly detection in energy systems, laying the groundwork for future deployment in grid monitoring platforms and decision-support tools for market operators and policymakers.

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Appendix 1: Github Link

<https://github.com/VarunJaiswal05/electricity-price-anomaly-detection-QLD.git>

Appendix 2: Supplementary Information

Apendix

June 1, 2025

```
[ ]: # Load Required Libraries
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from sklearn.ensemble import IsolationForest
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, LSTM, RepeatVector, TimeDistributed, Dense
import matplotlib.pyplot as plt

# Load the Dataset
df = pd.read_csv("C:/Users/Varun/OneDrive/Desktop/DATASET/merged_cleaned_data.csv", parse_dates=['SETTLEMENTDATE'])

# Feature Engineering
# Ensure datetime conversion
df['SETTLEMENTDATE'] = pd.to_datetime(df['SETTLEMENTDATE'], errors='coerce')

# Create lagged price
df['lagged_RRP'] = df['RRP'].shift(1).bfill()

# Calculate percentage price change
df['price_change_pct'] = df['RRP'].pct_change().fillna(0) * 100

# Calculate demand forecast error
df['demand_error'] = df['TOTALDEMAND'] - df['DEMANDFORECAST']

# Extract temporal features
df['hour_of_day'] = df['SETTLEMENTDATE'].dt.hour
df['day_of_week'] = df['SETTLEMENTDATE'].dt.dayofweek

# Define Selected Features
features = [
    'RRP', 'lagged_RRP', 'price_change_pct', 'demand_error',
    'TOTALDEMAND', 'AVAILABLEGENERATION', 'DISPATCHABLEGENERATION',
    'TRADING_RRP', 'TEMPERATURE', 'SOLAR_RADIATION', 'RAINFALL',
    'hour_of_day', 'day_of_week'
```



```

]

[ ]: df

[ ]: df.columns

[ ]: # Drop NaNs, handle inf values, and scale the features
X = df[features].replace([np.inf, -np.inf], np.nan).dropna()
scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

[ ]: # -----
# Isolation Forest Anomaly Detection
# -----
# Train Isolation Forest on full scaled feature set
iso_model = IsolationForest(n_estimators=100, contamination=0.02,
                             random_state=42)
df_iso = X.copy()
df_iso['iso_anomaly'] = iso_model.fit_predict(X_scaled)
# Convert output: -1 = anomaly + 1, 1 = normal + 0
df_iso['iso_anomaly'] = df_iso['iso_anomaly'].map({1: 0, -1: 1})

[ ]: df_iso

[ ]: df_iso.info()

[ ]: # Count and print number of anomalies detected by Isolation Forest
num_iso_anomalies = df_iso['iso_anomaly'].sum()/len(df_iso)
print(f"Number of anomalies detected by Isolation Forest: {num_iso_anomalies}")

[ ]: # Create Sliding Windows for LSTM
window_size = 96
X_seq = []
for i in range(len(X_scaled) - window_size):
    X_seq.append(X_scaled[i:i+window_size])
X_seq = np.array(X_seq)

# Build LSTM Autoencoder Model
input_dim = X_seq.shape[2]
inputs = Input(shape=(window_size, input_dim))
encoded = LSTM(64, return_sequences=False)(inputs)
bottleneck = RepeatVector(window_size)(encoded)
decoded = LSTM(64, return_sequences=True)(bottleneck)
outputs = TimeDistributed(Dense(input_dim))(decoded)

model = Model(inputs, outputs)
model.compile(optimizer='adam', loss='mse')

```

```

# Train the LSTM Autoencoder
model.fit(X_seq, X_seq, epochs=30, batch_size=64, validation_split=0.1,
         verbose=1)

[ ]: # Reconstruct and Compute Reconstruction Error
X_pred = model.predict(X_seq)
reconstruction_errors = np.mean(np.square(X_seq - X_pred), axis=(1, 2))

# Set Threshold and Flag Anomalies
threshold = np.percentile(reconstruction_errors, 95)
anomalies = reconstruction_errors > threshold

# Attach LSTM Anomaly Labels to Data
anomaly_df = X.iloc>window_size:].copy()
anomaly_df['reconstruction_error'] = reconstruction_errors
anomaly_df['lstm_anomaly'] = anomalies.astype(int)

# Combine Both Anomaly Flags
anomaly_df['iso_anomaly'] = df_iso.iloc>window_size:]['iso_anomaly'].values

[ ]: anomaly_df.tail(10)

[ ]: # Print anomaly percentages from LSTM and Isolation Forest
lstm_count = anomaly_df['lstm_anomaly'].sum()
lstm_percent = lstm_count / len(anomaly_df) * 100

iso_count = anomaly_df['iso_anomaly'].sum()
iso_percent = iso_count / len(anomaly_df) * 100

print(f"LSTM Anomalies: {lstm_count} ({lstm_percent:.2f}%)")
print(f"Isolation Forest Anomalies: {iso_count} ({iso_percent:.2f}%)")

[ ]: # Compare agreement between LSTM and Isolation Forest
anomaly_df['both_agree'] = (anomaly_df['lstm_anomaly'] == 1) &
    (anomaly_df['iso_anomaly'] == 1)
both_agree_count = anomaly_df['both_agree'].sum()
print(f"Both models agree on {both_agree_count} anomalies ({(both_agree_count/
    len(anomaly_df))*100:.2f}%)")

[ ]: # Save final combined anomaly output for future use
anomaly_df.to_csv("anomaly_detected_combined.csv", index=False)

[ ]: from matplotlib_venn import venn2
from matplotlib import pyplot as plt

plt.figure(figsize=(6,6))

```

```
venn2(subsets = (
    ((anomaly_df['lstm_anomaly'] == 1) & (anomaly_df['iso_anomaly'] == 0)).
    sum(),
    ((anomaly_df['iso_anomaly'] == 1) & (anomaly_df['lstm_anomaly'] == 0)).
    sum(),
    ((anomaly_df['lstm_anomaly'] == 1) & (anomaly_df['iso_anomaly'] == 1)).sum()
), set_labels=('LSTM', 'Isolation Forest'))
plt.title("Overlap of Detected Anomalies")
plt.show()
```

```
[ ]: # Top anomalies by LSTM reconstruction error
top_anomalies = anomaly_df.sort_values(by='reconstruction_error',
    ascending=False).head(10)
print(top_anomalies[['RRP', 'reconstruction_error', 'lstm_anomaly',
    'iso_anomaly']])
```

```
[ ]: # Check for RRP values equal to -999.99 in the original dataset
invalid_rrp_rows = df[df['RRP'] == -999.99]

# Display results
print(f"Number of rows with RRP = -999.99: {len(invalid_rrp_rows)}")
invalid_rrp_rows.head()
```

```
[ ]: # Load the dataset
df = pd.read_csv("anomaly_detected_combined.csv")

# Create 'both_agree' column if not present
df['both_agree'] = (df['lstm_anomaly'] == 1) & (df['iso_anomaly'] == 1)

# Create index for plotting (since no timestamp column)
df['index'] = range(len(df))

# Plot RRP time-series
plt.figure(figsize=(14, 6))
plt.plot(df['index'], df['RRP'], label='RRP', color='black', linewidth=1)

# Overlay anomalies
plt.scatter(df[df['lstm_anomaly'] == 1]['index'], df[df['lstm_anomaly'] == 1]
    ['RRP'],
    color='red', label='LSTM Anomaly', s=10)
plt.scatter(df[df['iso_anomaly'] == 1]['index'], df[df['iso_anomaly'] == 1]
    ['RRP'],
    color='blue', label='IF Anomaly', s=10)
plt.scatter(df[df['both_agree'] == True]['index'], df[df['both_agree'] == True]
    ['RRP'],
    color='purple', label='Both Models', s=20, edgecolor='white')
```

```
# Formatting
plt.title("Time-Series of RRP with Anomalies Detected by LSTM and IF")
plt.xlabel("Index (Chronological Order)")
plt.ylabel("Regional Reference Price (RRP)")
plt.legend()
plt.tight_layout()
plt.show()
```

```
[ ]: # Load your data
df = pd.read_csv("anomaly_detected_combined.csv")

# Confirm both_agree column exists
df['both_agree'] = (df['lstm_anomaly'] == 1) & (df['iso_anomaly'] == 1)

# Keep only original (unscaled) meaningful features
core_features = [
    'RRP', 'lagged_RRP', 'price_change_pct', 'demand_error', 'TOTALDEMAND',
    'AVAILABLEGENERATION', 'DISPATCHABLEGENERATION', 'TRADING_RRP',
    'TEMPERATURE', 'SOLAR_RADIATION', 'RAINFALL'
]

# Calculate median deviation
normal_median = df[df['both_agree'] == False][core_features].median()
anomaly_median = df[df['both_agree'] == True][core_features].median()
deviation = (anomaly_median - normal_median).abs().sort_values(ascending=False)

# Plot top 10
plt.figure(figsize=(10, 5))
deviation.head(10).plot(kind='bar', color='steelblue')
plt.title("Top 10 Median Deviations in Common Anomalies")
plt.ylabel("Absolute Median Deviation")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
[ ]: import pandas as pd
import matplotlib.pyplot as plt

# Load your anomaly detection results (no date column to parse)
df = pd.read_csv("anomaly_detected_combined.csv")

# Create the both_agree column
df['both_agree'] = (df['lstm_anomaly'] == 1) & (df['iso_anomaly'] == 1)

# Filter where both models agree
common_anomalies = df[df['both_agree'] == True]
```

```
# Sort by reconstruction error and pick top 10
top_common = common_anomalies.sort_values(by='reconstruction_error',
    ascending=False).head(10)

# Display the top anomalies
print("Top 10 High-Confidence Anomalies (LSTM + IF):")
print(top_common[['RRP', 'reconstruction_error', 'TOTALDEMAND', 'TEMPERATURE',
    'demand_error']])

# Visualize as a bar chart
plt.figure(figsize=(10, 5))
plt.barh(top_common.index.astype(str), top_common['reconstruction_error'],
    color='red')
plt.xlabel("Reconstruction Error")
plt.title("Top 10 Common Anomalies Detected by Both Models")
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```

```
[ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load anomaly file
df = pd.read_csv("anomaly_detected_combined.csv")

# Create time-based features if not already present
df['hour_of_day'] = df.get('hour_of_day', pd.Series(range(len(df))) % 24)
df['day_of_week'] = df.get('day_of_week', pd.Series(range(len(df))) % 7)

# Create a pivot table showing count of LSTM anomalies by hour and day
heatmap_data = df[df['lstm_anomaly'] == 1].pivot_table(
    index='day_of_week',
    columns='hour_of_day',
    values='RRP',
    aggfunc='count',
    fill_value=0
)

# Plot the heatmap
plt.figure(figsize=(12, 6))
sns.heatmap(heatmap_data, cmap="Reds", annot=True, fmt='d')
plt.title("Heatmap of LSTM Anomalies by Day of Week and Hour of Day")
plt.xlabel("Hour of Day")
plt.ylabel("Day of Week (0 = Monday)")
plt.tight_layout()
plt.show()
```



```
[ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load your data
df = pd.read_csv("anomaly_detected_combined.csv")

# Scatter plot: RRP vs Temperature, colored by LSTM anomaly
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='TEMPERATURE', y='RRP', hue='lstm_anomaly',
                palette={0: 'gray', 1: 'red'}, alpha=0.7)

plt.title("RRP vs Temperature with LSTM Anomalies Highlighted")
plt.xlabel("Temperature (°C)")
plt.ylabel("Regional Reference Price (RRP)")
plt.legend(title="LSTM Anomaly", labels=["Normal", "Anomaly"])
plt.tight_layout()
plt.show()
```

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